SATYAM SHARMA

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RESEARCH INTERESTS

Reinforcement learning when feedback is rare or delayed. My research asks what an agent should remember from its own experience, and how it can discover which of its past events actually caused success, without anyone telling it what matters. I want to carry these questions into **adaptive decision-making systems and LLM-based agents**, with experiments careful enough to tell a genuine gain apart from a benchmark artifact.

Availability: based in Tokyo; available part-time during term and full-time during university breaks.

EDUCATION

Institute of Science Tokyo (formerly Tokyo Institute of Technology) Apr 2026 – Mar 2028 (expected)
M.S., Industrial Engineering and Economics (Machine Learning and Reinforcement Learning) *GPA 3.83 / 4.5*

- Ichise Laboratory (Prof. Ryutaro Ichise); research student in the same lab from Oct 2025 to Mar 2026.
- Coursework: Intelligent Systems Engineering; From Data Analytics to Machine Learning.

Sant Longowal Institute of Engineering and Technology (SLIET), India Jul 2021 – Jun 2025
B.E. (Honours), Computer Science and Engineering *CGPA 8.9 / 10*

- Honours track in Optimization Techniques and Data Analytics.

HONORS AND AWARDS

MEXT (Monbukagakusho) Scholarship, Government of Japan Oct 2025 – Present

- **One of 25 Indian nationals** selected in 2025; fully funded master's study and research at the Institute of Science Tokyo.

Institute Project Funding (INR 50,000), SLIET Undergraduate

- Awarded for Earth Docker; the **only second-year team selected**, alongside final-year students.

RESEARCH EXPERIENCE

Graduate Researcher, Ichise Laboratory, Institute of Science Tokyo Oct 2025 – Present
Thesis: memory and event discovery for sparse-reward reinforcement learning. Advisor: Prof. Ryutaro Ichise

- Studying **what an agent should remember when reward is rare** and nobody specifies what matters. Built a modular PPO research codebase (Python, PyTorch) in which knowledge-graph, flat episodic and count-based memories plug into one **potential-based reward-shaping** interface, so they are compared fairly without changing the optimal policy.
- Ran **~2,400 seeded runs** on MiniGrid: a memory bonus raised DoorKey-8x8 from **3 to 22+ of 24** successful seeds, and a hand-written graph memory beat flat memory on DoorKey-6x6 (48 seeds, p = 0.0026) but not elsewhere. This motivated letting the agent **discover structure instead of having it written by hand**.
- Designed a **domain-agnostic event detector**, with no knowledge of keys or doors, that confirms each candidate event by **counterfactual replay**. During ordinary training it found the **correct event chain in 48 of 48 runs** in each of five games and **invented none across 432 control runs**.

- Designed and ran a **pre-registered benchmark of 4,308 runs** on two shared lab servers (3,552 scored runs over 11 games and 7 methods with 48 paired seeds, plus tuning; Holm-corrected tests). The discovered chain **roughly halved PPO's training time** on UnlockPickup and ObstructedMaze (faster on 47 of 48 seeds), and a perfect chain made it up to **7.5x faster**.
- Reported the negative half with the same rigor: **flat memory was faster in 6 of 8 games**. The gap to the perfect-chain ceiling points to a testable cause (the chain arrives late), now being checked with analyses declared before any result.
- Showed with a test that the public code of **VACERL (ECML PKDD 2024)**, the closest prior method, never trains its learner on its own discovered bonus.
- **Retracted my own headline result** after a failed reproducibility check exposed an episode-boundary bug worth ~24,000 steps of false advantage; fixed and re-validated it on fresh seeds. In the latest benchmark, **two independent scoring programs agreed on every number**.

PROJECTS

EduMind AI, LLM exam-preparation tutor Code Demo 2025

- Built a tutor for **8 Indian competitive exams** (UPSC, GATE, NEET and others) that runs entirely on a student's laptop: locally served **LLaMA 3.1/3.2** (Ollama) behind a React/TypeScript and Node.js stack, covering Q&A, practice tests and quizzes.
- Added **essay evaluation with structured feedback** and a 30-minute cache that skips repeat model calls.

Reel Vault, budget-aware multimodal LLM pipeline (personal tool, code private) 2026

- Turns saved Instagram reels into a searchable, mapped knowledge base: one "5 best ramen spots" post becomes five place cards pinned on a map.
- Extraction **escalates only when it must** (caption, then Whisper, then Gemini vision on frames), with daily caps checked before every paid call, so it **runs entirely on free tiers**, with **293 automated tests**.

Earth Docker, autonomous agricultural robot Video Undergraduate

- Field robot for early pest detection: trained a **ResNet152V2 to 98% accuracy** on diseased vs healthy cotton plants; with a robotics-club team, ran perception and **RP-LiDAR navigation on a Raspberry Pi 4**.

TECHNICAL SKILLS

Reinforcement learning: PPO, DQN, SAC, reward shaping, Stable-Baselines3, Gymnasium, MiniGrid

Machine learning: PyTorch, NumPy, Pandas, scikit-learn, TensorFlow / Keras

LLMs: Ollama, LLaMA 3, Gemini API, Whisper; cost-aware LLM pipelines; learning post-training (SFT, GRPO)

Research methods: Pre-registration, paired-seed tests (Holm), independent re-scoring, Linux servers, MLflow

Software: Python (primary), TypeScript, C/C++; FastAPI, React, Node.js, SQLite; Git; Docker (basic)

Languages: English (fluent), Japanese (basic, around JLPT N4)

LEADERSHIP

Convenor, Robotics Club, SLIET Feb 2022 – Jul 2025

- Led drone, line-following robot and hovercraft projects; represented the club at IIT-BHU Technex 2023 and coordinated the inter-college Robozar events at TechFest 2023.